BJT small-signal I

Cheat sheet · Day 10 · Mon 2 Nov · pack E6

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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BJT SMALL-SIGNAL I  (VT = 26 mV Boylestad / 25 mV Sedra - state which)
gm = IC/VT     r_pi = beta/gm = VT/IB = (beta+1) re ~ beta re
re = VT/IE = alpha/gm     ro = VA/IC = 1/hoe     gm*re = alpha ~ 1
Models: hybrid-pi (r_pi, gm v_pi, ro) == re model (beta re, beta Ib, ro)
        since gm v_pi = gm r_pi ib = beta ib.   T model: re in emitter, alpha ie C->E
h-params (CE): hie = beta re, hfe = beta, hoe = 1/ro, hre ~ 0
Small-signal valid: |vbe| < ~10 mV

AC EQUIVALENT: DC sources -> ground, all C -> short, delete shorted R, redraw, insert

ALWAYS: 1) DC Q-point (caps open) -> IE   2) re = 26mV/IE   3) AC circuit   4) Zi, Zo
CE FIXED BIAS:   IB=(VCC-0.7)/RB, IE=(b+1)IB
   Zi = RB || beta re     Zo = RC || ro     Av = -(RC||ro)/re  (~ -RC/re if ro >= 10R
CE VDB, RE BYPASSED:  test beta RE >= 10 R2 -> VB = R2 VCC/(R1+R2), IE=(VB-0.7)/RE
                      else Thevenin: IB = (ETh-0.7)/(RTh+(b+1)RE)
   Zi = R1||R2||beta re   Zo = RC || ro     Av = -(RC||ro)/re
Phase: minus sign = 180 deg.   Check: |Av| = gm RC = V_RC/VT
Totals: vC(t) = VC + Av vbe(t);  iB(t) = IB + vbe/r_pi
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